

## **MANIPAL UNIVERSITY JAIPUR**

School of Computer Science and Engineering

Department of Data Science and Engineering

Course Hand-out

### **A. Basic Details:**

|                                                                                                             |                                                                           |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Programme Name:                                                                                             | B. Tech. (Data Science and Engineering)                                   |
| Course Name:                                                                                                | Natural Language Processing                                               |
| Course Code:                                                                                                | DSE 4150                                                                  |
| LTPC ( <i>Lecture Tutorial Practical Credits</i> ):                                                         | 2-1-0-3                                                                   |
| Session:                                                                                                    | July-November 2026                                                        |
| Class:                                                                                                      | 7 <sup>th</sup> Semester                                                  |
| Course Coordinator:                                                                                         | Dr. Neha V Sharma                                                         |
| Course Instructor(s):                                                                                       | Dr. Neha V. Sharma, Dr. Pooja Gupta, Dr. Jayanthi                         |
| Additional Practitioner(s) – if any<br>( <i>Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.</i> ): | To be identified and appointed later/ or at the beginning of the semester |

**B. Introduction:** This course is offered for Data Science Students as a core subject, targeting students who wish to pursue development in industries or higher studies in field of Natural Language Processing and allied areas. This course aims to make the students understand the models, methods, and algorithms of Natural Language Processing for common NLP tasks, such as speech recognition, machine translation, spam filtering, text classification, spell checking etc. After learning through this course, students will be able to understand and implement probabilistic models, estimate parameters for such models, and run meaningful experiments to validate such language models. The student will gain understanding of linguistic phenomena and will explore the linguistic features relevant to each NLP task. Prerequisite for the course is knowledge of fundamental mathematics including Linear algebra Probability and Statistics, AI, and programming in any high-level language, preferably python.

**C. Course Outcomes:**

| <b>CO Statement</b>                                                                                                          | <b>CO</b> | <b>Level</b>       | <b>Target Attainment %</b> | <b>Target Attainment level</b> |
|------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------|----------------------------|--------------------------------|
| Remember and identify fundamental terms and concepts in Natural Language Processing.                                         | DSE4150.1 | L2<br>(Understand) | 70                         | 2                              |
| Understand FSM and language morphology with FSTs.                                                                            | DSE4150.2 | L2<br>(Understand) | 80                         | 3                              |
| Study text processing and language modeling to understand the relation between speech and grammatical structure of language. | DSE4150.3 | L3<br>(Apply)      | 75                         | 2                              |
| Explore the part of speech tagging and syntactic analysis in terms of natural language expression.                           | DSE4150.4 | L3<br>(Apply)      | 85                         | 3                              |
| Design and implement NLP techniques to real-world applications.                                                              | DSE4150.5 | L6<br>(Create)     | 75                         | 2                              |

Here, the level indicates the cognitive level of Revised Bloom's Taxonomy.

Information about attainment levels:

| <b>Attainment (%)</b> | <b>Level</b> |
|-----------------------|--------------|
| < 60 %                | 0            |
| $\geq 60\% < 70\%$    | 1            |
| $\geq 70\% < 80\%$    | 2            |
| $\geq 80$             | 3            |

**D. Program Outcomes and Program Specific Outcomes:**

**[PO.1].** Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems

**[PO.2].** Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

**[PO.3].** Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

**[PO.4].** Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

**[PO.5].** Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations

**[PO.6].** The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice

**[PO.7].** Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

**[PO.8].** Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices

**[PO.9].** Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

**[PO.10].** Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

**[PO.11].** Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments

**[PSO.1].** Understand the role of mathematics, statistics, and AI techniques in the field of data science & engineering.

**[PSO.2].** Apply the acquired knowledge and expertise to perform data analytics tasks for multidimensional data sets.

**[PSO.3].** Develop effective and scalable industrial solutions for real world socio-economic problems using data analytics tools and techniques.

**E. Assessment Plan:**

| Criteria                           | Description                                                                                            | Maximum Marks |
|------------------------------------|--------------------------------------------------------------------------------------------------------|---------------|
| Internal Assessment<br>(Summative) | Mid-Term Examination<br>(Close Book)                                                                   | 30            |
|                                    | Class Work Sessional (CWS):<br>Quiz, Assignment, Project,<br>Research Paper, Video<br>Assignment, etc. | 30            |
| End Term Exam<br>(Summative)       | End Term Examination<br>(Close)                                                                        | 40            |
|                                    | Total                                                                                                  | 100           |

**F. Syllabus:** Basics of Finite State Automata, Knowledge in Speech and Language Processing, Ambiguity, Models and Algorithm, Survey of English Morphology, Finite-State Morphological Parsing, Building a Finite State Lexicon, FSTs for Morphological Parsing, Lexicon-Free FSTs, Words and sentence tokenization, Detecting and Correcting Spelling Errors, Case study: Normalizing Text, Segmentation. N-Grams, Unsmoothed N-Grams, Smoothing, English Word Classes, Tag- sets for English, Part-of-Speech Tagging, Case study: Automatic Tagging. Constituency, Some Grammar Rules for English, The Penn Treebank project, Dependency Grammar, Parsing with Context Free Grammars, CKY algorithm.

**G. Text Books:**

**T1.** D. Jurafsky, J. H. Martin, “Speech and Language Processing”, 2e, Pearson Education, 2009 (E-book Link: Speech and Language Processing (stanford.edu))

**T2.** Akshar Bharati, Rajeev Sangal and Vineet Chaitanya, Natural Language Processing: A Paninian Perspective, Prentice-Hall of India, New Delhi, 1995.

**Reference Books:**

**R1.** C. Eugene, “Statistical Language Learning”, MIT Press, 1999.

**R2.** T. Siddiqui, U. S. Tiwary, “Natural language processing and Information retrieval”, OUP, 2008.

**R3.** Chris Manning, Hinrich Schutze, Foundations of Statistical Natural Language Processing, MIT Press, Cambridge, 1999.

### Lecture Plan:

| Lecture No | Major Topics               | Topics                                                                           | Mode of Delivery                 | Corresponding CO | Mode of Assessing the Outcome |
|------------|----------------------------|----------------------------------------------------------------------------------|----------------------------------|------------------|-------------------------------|
| 2          | Introduction               | Phases of NLP                                                                    | Lecture based teaching-learning  | DSE4150.1        | Mid Term, End Term, CWS       |
| 3          | Introduction               | Knowledge in Speech and Language Processing and Ambiguity                        | Learning through Problem Solving | DSE4150.1        | Mid Term, End Term, CWS       |
| 4          | Introduction               | Knowledge in Speech and Language Processing and Ambiguity                        | Lecture based teaching-learning  | DSE4150.1        |                               |
| 5          | Introduction               | Overview of different models and algorithms in NLP                               | Lecture based teaching-learning  | DSE4150.1        | Mid Term, End Term, CWS       |
| 6          | Finite State Machine (FSM) | Basics of Finite State Automata – Acceptor and Transducer and its Categorization | Learning through Problem Solving | DSE4150.2        | Mid Term, End Term, CWS       |
| 7          | Finite State Machine (FSM) | Formal language and its acceptance, DFA and NFA                                  | Lecture based teaching-learning  | DSE4150.2        | Mid Term, End Term, CWS       |
| 8          | Finite State Machine (FSM) | DFA                                                                              | Lecture based teaching-learning  | DSE4150.2        |                               |
| 9          | Finite State Machine (FSM) | NFA                                                                              | Lecture based teaching-learning  | DSE4150.2        |                               |
| 10         | Finite State Machine (FSM) | Relation and difference between DFA and NFA, Basic Regular Expression Patterns   | Learning through Problem Solving | DSE4150.2        | Mid Term, End Term, CWS       |
| 11         | Morphological Analysis     | Survey of English Morphology                                                     | Lecture based teaching-learning  | DSE4150.2        | Mid Term, End Term, CWS       |
| 12         | Morphological Analysis     | Inflectional Morphology                                                          | Lecture based teaching-learning  | DSE4150.3        | Mid Term, End Term, CWS       |
| 13         | Morphological Analysis     | Derivational Morphology                                                          | Lecture based teaching-learning  | DSE4150.3        |                               |
| 14         | Morphological Analysis     | Finite-State Morphological Parsing                                               | Lecture based teaching-learning  | DSE4150.3        | Mid Term, End Term, CWS       |

|    |                               |                                          |                                     |           |                               |
|----|-------------------------------|------------------------------------------|-------------------------------------|-----------|-------------------------------|
| 15 | <b>Morphological Analysis</b> | Building a Finite State Lexicon          |                                     | DSE4150.3 |                               |
| 16 | <b>Morphological Analysis</b> | FSTs for Morphological Parsing           | Individual Learning/<br>Self Study  | DSE4150.3 | Mid Term,<br>End Term,<br>CWS |
| 17 | <b>Morphological Analysis</b> | Orthographic Rules                       | Lecture based<br>teaching-learning  | DSE4150.3 |                               |
| 18 | <b>Morphological Analysis</b> | Lexicon -Free FSTs                       | Lecture based<br>teaching-learning  | DSE4150.3 | Mid Term,<br>End Term,<br>CWS |
| 19 | <b>Morphological Analysis</b> | Inflectional & Derivational Morphology   | Lecture based<br>teaching-learning  | DSE4150.3 | Mid Term,<br>End Term,<br>CWS |
| 20 | <b>Text Processing</b>        | Words and sentence tokenization          | Lecture based<br>teaching-learning  | DSE4150.4 | Mid Term,<br>End Term,<br>CWS |
| 21 | <b>Text Processing</b>        | Detecting and Correcting Spelling Errors | Learning through<br>Problem Solving | DSE4150.4 | Mid Term,<br>End Term,<br>CWS |
| 22 | <b>Text Processing</b>        | Case Study: Normalizing Text             | Lecture based<br>teachin-learning   | DSE4150.4 | Mid Term,<br>End Term,<br>CWS |
| 23 | <b>Text Processing</b>        | Segmentation                             | Lecture based<br>teachin-learning   | DSE4150.4 | Mid Term,<br>End Term,<br>CWS |
| 24 | <b>Text Processing</b>        | Correcting Spelling Errors               | Learning through<br>Problem Solving | DSE4150.4 | Mid Term,<br>End Term,<br>CWS |
| 25 | <b>Language Modeling</b>      | N-Grams                                  | Lecture based<br>teachin-learning   | DSE4150.4 | Mid Term,<br>End Term,<br>CWS |
| 26 | <b>Language Modeling</b>      | Simple or Unsmoothed N-Grams             | Lecture based<br>teachin-learning   | DSE4150.5 | Mid Term,<br>End Term,<br>CWS |
| 27 | <b>Language Modeling</b>      | Smoothing Techniques                     | Learning through<br>Problem Solving | DSE4150.5 | Mid Term,<br>End Term,<br>CWS |
| 28 | <b>Part-of-Speech Tagging</b> | English Word Classes                     | Lecture based<br>teachin-learning   | DSE4150.5 | Mid Term,<br>End Term,<br>CWS |
| 29 | <b>Part-of-Speech Tagging</b> | Tagsets for English                      | Lecture based<br>teachin-learning   | DSE4150.5 | Mid Term,<br>End Term,<br>CWS |
| 30 | <b>Part-of-Speech Tagging</b> | Part-of-Speech Tagging (HMM) Algorithms  | Lecture based<br>teachin-learning   | DSE4150.5 | Mid Term,<br>End Term,<br>CWS |
| 31 | <b>Part-of-Speech Tagging</b> | Rule-based Part-of-Speech Tagging        | Learning through<br>Problem Solving | DSE4150.5 | Mid Term,<br>End Term,<br>CWS |

|    |                                       |                                      |                                  |           |                         |
|----|---------------------------------------|--------------------------------------|----------------------------------|-----------|-------------------------|
| 32 | <b>Syntactic Analysis</b>             | Overview of Constituency Parsing     | Lecture based teaching-learning  | DSE4150.5 | Mid Term, End Term, CWS |
| 33 | <b>Syntactic Analysis</b>             | Grammar Rules for English            | Group based teaching learning    | DSE4150.5 | Mid Term, End Term, CWS |
| 34 | <b>Syntactic Analysis</b>             | Dependency Grammar                   | Learning through Problem Solving | DSE4150.5 | Mid Term, End Term, CWS |
| 35 | Machine Learning: Classification      | Parsing with Context-Free Grammars   | Lecture based teaching-learning  | DSE4150.5 | End Term, CWS           |
| 36 | Machine Learning: Classification Tree | Case Study: Automatic Tagging        | Lecture based teaching-learning  | DSE4150.5 | End Term, CWS           |
| 37 | Machine Learning: Classification Tree | Case Study: The Pan Treebank Project | Learning through Problem Solving | DSE4150.5 | End Term, CWS           |
| 38 | Machine Learning: Classification Tree | Case Study: Sentiment Analysis       | Lecture based teaching-learning  | DSE4150.5 | End Term, CWS           |
| 39 | Machine Learning: Classification Tree | CKY Algorithm                        | Flipped Classroom                | DSE4150.5 | End Term, CWS           |

### Outside-Class Engagement Activities

*To complete the required notional learning hours as per NCeF guidelines*

| Type of Activity                                                                                                    | Difficulty Level | Hours Engaged   |
|---------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| Watching recommended video lectures and tutorials on course topics                                                  | Easy             | 10              |
| Case Study / Reasoning based Assignments                                                                            | Moderate         | 14              |
| Developing a mini-project or prototype using data analysis and performing regression and classification based tasks | Difficult        | 30              |
| <b>Total Engagement Hours</b>                                                                                       |                  | <b>54 Hours</b> |

## H. Course Articulation Matrix:

| CO         | STATEMENT                                                                                                                    | CORRELATION WITH PROGRAM OUTCOMES |      |      |      |      |      |      |      |      |       |       | CORRELATION WITH PROGRAM SPECIFIC OUTCOMES |       |       |
|------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------|------|------|------|------|------|------|------|-------|-------|--------------------------------------------|-------|-------|
|            |                                                                                                                              | PO 1                              | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 | PO 11 | PSO 1                                      | PSO 2 | PSO 3 |
| DSE415 0.1 | Remember and identify fundamental terms and concepts in Natural Language Processing.                                         | 2                                 | 3    |      |      |      |      | 1    | 1    |      | 1     |       | 1                                          | 2     |       |
| DSE415 0.2 | Understand FSM and language morphology with FSTs.                                                                            | 1                                 | 2    | 2    |      |      | 3    | 1    | 1    |      | 1     | 2     | 2                                          | 1     | 3     |
| DSE415 0.3 | Study text processing and language modeling to understand the relation between speech and grammatical structure of language. |                                   | 2    |      | 2    | 2    |      | 1    | 1    |      |       | 1     | 2                                          | 1     |       |
| DSE415 0.4 | Explore the part of speech tagging and syntactic analysis in terms of natural language expression.                           |                                   | 2    | 2    | 2    | 2    | 2    | 1    | 2    | 3    |       |       | 2                                          | 1     | 3     |
| DSE415 0.5 | Design and implement NLP techniques to real-world applications.                                                              | 3                                 |      | 2    |      | 1    | 1    |      | 1    |      | 2     |       | 1                                          | 3     |       |

0-No Attainment; 1- Low Attainment; 2- Moderate Attainment; 3- Substantial Attainment

